"For to all those who have, more will be given, and they will have an abundance; but from those who have nothing, even what they have will be taken away."

MATTHEW 25:29 (Holy Bible, New Revised Standard Version, 1989)

Emergent Inequality from Equal Starts:

Agent-Based Wealth Exchange and the Matthew Effect

Working Paper

Agent-Based Computational Economics

July 3, 2026

Abstract

How does substantial inequality arise when economic agents begin with identical endowments and face no inherent differences in ability or preference? This paper develops an agent-based computational model of stochastic wealth exchange in a closed economy and shows that inequality is an *emergent* property of interaction dynamics, not a prerequisite of the initial state. We formalize four canonical exchange rules from the econophysics and agent-based literatures: random fixed transfers (Dragulescu and Yakovenko, 2000), saving-propensity exchange (Chakraborti and Chakrabarti, 2000), proportional yard-sale losses (Ispolatov et al., 1998), and a minimal Matthew-effect specification in which wealth flows preferentially toward the richer party in each encounter (Merton, 1968; Price, 1976). Despite conservation of aggregate wealth and symmetric initial conditions, all specifications generate dispersion; some produce extreme *wealth condensation* in which a vanishing fraction of agents holds a positive fraction of total resources (Bouchaud and Mézard, 2000; Yakovenko and Rosser, 2009). We report simulation evidence on the Gini coefficient, Lorenz curves, and top wealth shares, and interpret the results for readers without specialist training while preserving formal definitions. The exercise clarifies when "the rich get richer" follows from cumulative advantage embedded in mundane market-like encounters, and when it requires additional structural features such as heterogeneous returns or political institutions (Piketty, 2014; Acemoglu and Robinson, 2015).

Keywords: agent-based model; wealth distribution; Matthew effect; cumulative advantage; econophysics; emergent inequality; Gini coefficient
JEL codes: C63, D31, E17, C15

1 Introduction

A familiar puzzle runs through popular and academic discussions of inequality alike: if people start on roughly equal footing, why do outcomes diverge so sharply? The Gospel of Matthew offers a stark formulation of cumulative advantage—resources accrue to those who already hold them—and the sociologist Robert K. Merton imported the phrase *Matthew effect* into science studies to describe how small early advantages in recognition compound into large disparities in credit and resources (Merton, 1968). The underlying mechanism, *cumulative advantage* or *preferential attachment*, appears independently in bibliometrics (Price, 1976), network science, and models of asset exchange (Ispolatov et al., 1998; Bouchaud and Mézard, 2000).

This paper asks a disciplined version of the puzzle. Suppose N agents each begin with the same wealth $w_0 > 0$. They meet in pairs at random and transfer resources according to a simple rule that treats agents symmetrically *ex ante* (no agent is labeled "talented" or "high productivity" in the code). Aggregate wealth is conserved: what one agent loses, another gains. Can inequality still emerge?

The answer, in the models studied here, is yes. Inequality in these simulations is not assumed at $t = 0$; it *emerges* from the stochastic dynamics of exchange. That distinction matters methodologically. A high Gini coefficient in cross-sectional data does not by itself reveal whether inequality reflects initial heterogeneity, institutional rules, or the amplification of luck through interaction (Benhabib and Bisin, 2011; Galor and Zeira, 1993). Agent-based models (ABMs) make the mechanism visible: one can watch an economy depart from perfect equality even when every agent starts identical (Schelling, 1971; Epstein, 2006; Farmer and Foley, 2009).

We proceed in four steps. Section 2 defines key terms for a mixed audience of specialists and non-specialists. Section 3 presents the mathematical framework. Section 4 describes the computational experiment and figures. Section 5 connects results to economics and cautions against over-interpretation. Section 6 concludes.

2 Conceptual framework

2.1 Agent-based economic simulation

An **agent-based economic simulation** is a computer model in which many autonomous "agents" (households, traders, firms) follow explicit rules for interaction. The macroscopic patterns—here, the wealth distribution—are *generated* bottom-up from micro behavior rather than imposed by a representative agent or a closed-form equilibrium (Epstein, 2006; Farmer and Foley, 2009). This generative perspective is useful when the object of interest is the *distribution itself* and how it evolves over time.

2.2 Emergent inequality and wealth condensation

Emergent inequality refers to dispersion in outcomes that arises from the dynamics of the system even when initial conditions are symmetric. In the simulations below, all agents begin with $w_i(0) = w_0$. Any subsequent variance in $\{w_i(t)\}$ is entirely endogenous.

Wealth condensation is the extreme case in which, in the long run, a single agent (or a measure-zero set) holds a positive share of total wealth while most agents become impoverished (Bouchaud and Mézard, 2000; Ispolatov et al., 1998). Physicists describe this as a symmetry-breaking phase transition in exchange models with multiplicative updates. Economists might speak of "winner-take-all" dynamics. Both languages point to the same phenomenon: multiplicative feedback can convert microscopic randomness into macroscopic concentration.

2.3 The Matthew effect

Merton's *Matthew effect* names the social process by which "the rich get richer" in reputation, citations, or resources (Merton, 1968). Operationally, cumulative advantage requires that current stock $w_i(t)$ raise the expected increment $\mathbb{E}[\Delta w_i \mid w_i(t)]$. Our fourth model implements the minimal bias consistent with that idea: when two agents meet, wealth flows toward the richer party with probability $p > \frac{1}{2}$. No other heterogeneity is introduced.

Remark 1 (For the non-specialist reader)**.** *The **Gini coefficient** is a single number summarizing how unequally wealth is shared, ranging from* 0 *(everyone equal) to* 1 *(one person holds everything). The **Lorenz curve** plots the cumulative share of wealth held by the poorest* x*% of the population; the farther it bows below the 45-degree line of equality, the greater the inequality (Lorenz, 1905; Gini, 1912).*

3 Model

3.1 Environment

Consider a closed economy with a finite set of agents $\mathcal{N} = \{1, \ldots, N\}$. Time is discrete, $t = 0, 1, 2, \ldots$. Each agent i holds wealth $w_i(t) \geq 0$. Initial endowments are identical:

$$w_i(0) = w_0 \qquad \forall i \in \mathcal{N}. \tag{1}$$

Wealth is conserved at every step:

$$\sum_{i=1}^{N} w_i(t) = N w_0 \qquad \forall t. \tag{2}$$

Each period, a pair (i, j) is drawn uniformly from $\mathcal{N} \times \mathcal{N}$ (with replacement) and an exchange rule updates (w_i, w_j).

This is a *kinetic exchange model* of the kind surveyed by Yakovenko and Rosser (2009) and Lux (2008): agents behave like molecules colliding and transferring energy, except the traded quantity is money or wealth. The models are deliberately stripped down—no labor supply, no production, no banks—to isolate the effect of exchange rules on the distribution.

3.2 Inequality measures

Let $\mathbf{w} = (w_1, \ldots, w_N)$ denote the wealth vector with total $W = \sum_i w_i$. Sort wealth in ascending order: $w_{(1)} \leq \cdots \leq w_{(N)}$.

Definition 1 (Gini coefficient)**.** *The Gini index is*

$$G(\mathbf{w}) = \frac{2\sum_{k=1}^{N} k\, w_{(k)}}{N\sum_{k=1}^{N} w_{(k)}} - \frac{N+1}{N}. \tag{3}$$

If all w_i are equal, $G = 0$; if one agent holds all wealth, $G \to 1$ as $N \to \infty$.

The **top s% wealth share** is the fraction of W held by the richest $\lceil sN/100 \rceil$ agents. The Lorenz curve $L(x)$ gives the cumulative wealth share of the poorest fraction $x \in [0, 1]$ of the population.

3.3 Exchange specifications

Each model is defined by one micro-step applied to a randomly selected pair.

Model 1: Random exchange (Dragulescu–Yakovenko). Fix a stake $\Delta > 0$. With equal probability, either i pays j or j pays i, provided the payer's wealth exceeds Δ:

$$\begin{aligned} w_i &\leftarrow w_i - \Delta, \quad w_j \leftarrow w_j + \Delta \quad (\text{with prob. } \tfrac{1}{2}), \\ w_i &\leftarrow w_i + \Delta, \quad w_j \leftarrow w_j - \Delta \quad (\text{with prob. } \tfrac{1}{2}). \end{aligned} \tag{4}$$

This is a random walk of wealth shares with reflecting boundary at zero. Dragulescu and Yakovenko (2000) show that, under continuous-time limits, the stationary distribution of money approaches an exponential (Boltzmann–Gibbs) form when Δ is small relative to mean wealth—yet finite-N simulations still exhibit substantial dispersion (Figure 1).

Model 2: Saving propensity (Chakraborti–Chakrabarti). Each agent saves a fraction $\lambda \in [0, 1)$ before random redistribution of the remainder. When i and j meet, let $\tilde{w}_k = \lambda w_k$ be saved wealth and exchange $(1-\lambda)(w_i + w_j)/2$ according to a fair coin flip (Chakraborti and Chakrabarti, 2000; Chatterjee et al., 2004):

$$w_i' = \lambda w_i + \epsilon_i, \qquad w_j' = \lambda w_j + \epsilon_j, \tag{5}$$

where (ϵ_i, ϵ_j) redistribute the transferable pool. For $\lambda > 0$, the distribution develops a Pareto-like tail; $\lambda = 0$ returns to Model 1. Figure 7 shows how terminal inequality rises with λ.

Model 3: Yard-sale exchange (Ispolatov–Krapivsky–Redner). The loser pays a fraction $\beta \in (0,1)$ of the *smaller* stake $\min(w_i, w_j)$ to the winner, who is chosen with probability $\frac{1}{2}$:

$$\Delta = \beta \min(w_i, w_j), \qquad w_{\text{loser}} \leftarrow w_{\text{loser}} - \Delta, \quad w_{\text{winner}} \leftarrow w_{\text{winner}} + \Delta. \tag{6}$$

Using the minimum stake ties the traded quantity to the poorer party's scale and generates stronger feedback toward concentration than proportional loss on the loser's wealth alone (Ispolatov et al., 1998; Bouchaud and Mézard, 2000). Repeated meetings produce heavy-tailed outcomes and, in the long run, approximate wealth condensation.

Model 4: Matthew-effect exchange. Fix bias $p \in (\frac{1}{2}, 1)$ and stake $\Delta > 0$. When pair (i, j) meets, let r be the richer and p the poorer agent. With probability p, wealth flows from poor to rich; with probability $1 - p$, it flows the other way:

$$\mathbb{P}\big(w_r \uparrow,\ w_p \downarrow \ \big|\ \text{meeting}\big) = p > \tfrac{1}{2}. \tag{7}$$

This is the minimal operationalization of Merton's cumulative advantage in a pairwise economy. When $p = \frac{1}{2}$, the rule collapses to fair random exchange; when $p \to 1$, condensation accelerates (Figure 4).

Proposition 1 (Conservation). *Under Models 1–4, provided transfers are feasible (non-negative wealth), $\sum_i w_i(t)$ is invariant in t.*

Proof. Each rule transfers a quantity $\Delta \geq 0$ or $\beta w_{\text{loser}} \geq 0$ from one agent to another without leakage. □

Remark 2 (What the models omit). *Real economies contain production, heterogeneous human capital, credit constraints, taxation, and marriage of capital with political power (Piketty, 2014; Benhabib et al., 2019). The ABMs here isolate exchange dynamics. Matching their quantitative predictions to data requires embedding them in richer structures (Acemoglu and Robinson, 2015).*

4 Computational experiment

4.1 Implementation

We simulate $N = 1{,}000$ agents with $w_i(0) = 1$ for all i. Each model runs for 5×10^5 pairwise meetings unless otherwise noted. Parameters: $\Delta = 0.01$ (Models 1 and 4), $\lambda = 0.9$ (Model 2), $\beta = 0.1$ (Model 3), $p = 0.55$ (Model 4). Random seeds are fixed for reproducibility. Code is available in the project repository (`simulation/models.py`).

Table 1 reports terminal summary statistics from one run.

Table 1: Terminal inequality statistics ($N = 1000$, 5×10^5 steps)

Model	Gini	Top 1% share	Top 10% share	Max / Median
Random exchange (DY)	0.17	0.02	0.15	1.9
Saving ($\lambda = 0.9$)	0.67	0.16	0.52	68.7
Yard-sale ($\beta = 0.1$)	0.81	0.14	0.66	256.0
Matthew effect ($p = 0.55$)	0.41	0.03	0.22	2.8

Note: Values from the reference simulation (seed 42, $N = 1000$, 5×10^5 meetings). Yard-sale and saving-propensity models generate the strongest tail concentrations; random exchange produces modest dispersion despite identical starts.

4.2 Results

Inequality rises even from equality. Figure 1 plots the Gini coefficient at intervals of 5,000 steps. All four models depart from $G = 0$. The yard-sale specification converges fastest toward extreme concentration; random exchange rises more gradually but steadily.

Distribution shapes differ qualitatively. Figure 2 displays complementary cumulative distribution functions (CCDFs) on log-log axes. The yard-sale model exhibits a heavy tail consistent with condensation (Gabaix, 2009). The saving-propensity model produces intermediate tail behavior predicted by Chatterjee et al. (2004). Fair random exchange yields a distribution closer to exponential, yet still far from equality.

Lorenz curves. Figure 3 compares cumulative wealth shares. The yard-sale curve hugs the upper-left corner: the poorest half of the population holds essentially no wealth in the terminal snapshot. The Lorenz ordering mirrors the Gini ranking in Table 1.

Matthew effect comparative statics. Figure 4 varies the bias p and stake Δ in Model 4. Inequality is increasing in both parameters near the symmetric benchmark $p = \frac{1}{2}$. Even a modest 5 percentage-point bias ($p = 0.55$) materially raises the Gini over 500,000 interactions.

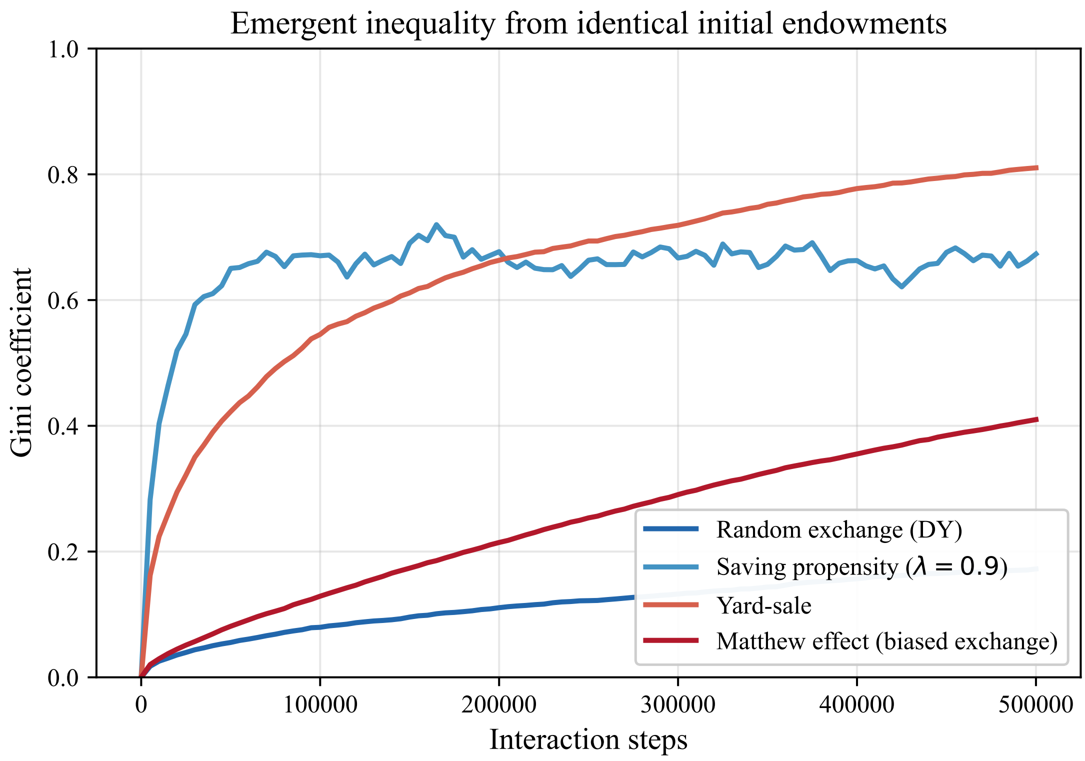

Figure 1: Evolution of the Gini coefficient under four exchange rules, starting from identical endowments ($w_i(0) = 1$).

Dynamics of top wealth shares. Figure 5 tracks the share held by the richest 1%. Yard-sale dynamics seize wealth rapidly; Matthew-effect and random-exchange models concentrate more slowly but monotonically.

From equality to skewness. Figure 6 shows wealth histograms for the Matthew-effect model at three snapshots. The distribution morphs from a spike at $w = 1$ into a right-skewed form without any change in fundamental agent characteristics—only the history of encounters differs across agents.

Role of saving. Figure 7 sweeps λ in Model 2. Higher saving propensity—holding more wealth fixed while trading the remainder—amplifies inequality, matching the analytical literature (Chakraborti and Chakrabarti, 2000; Yakovenko and Rosser, 2009).

5 Discussion

5.1 Mechanism and intuition

Three mechanisms recur across specifications.

First, **random walks with bounds**: when wealth can only be lost up to current holdings, agents near zero struggle to recover large losses. Symmetric coin flips nonethe-

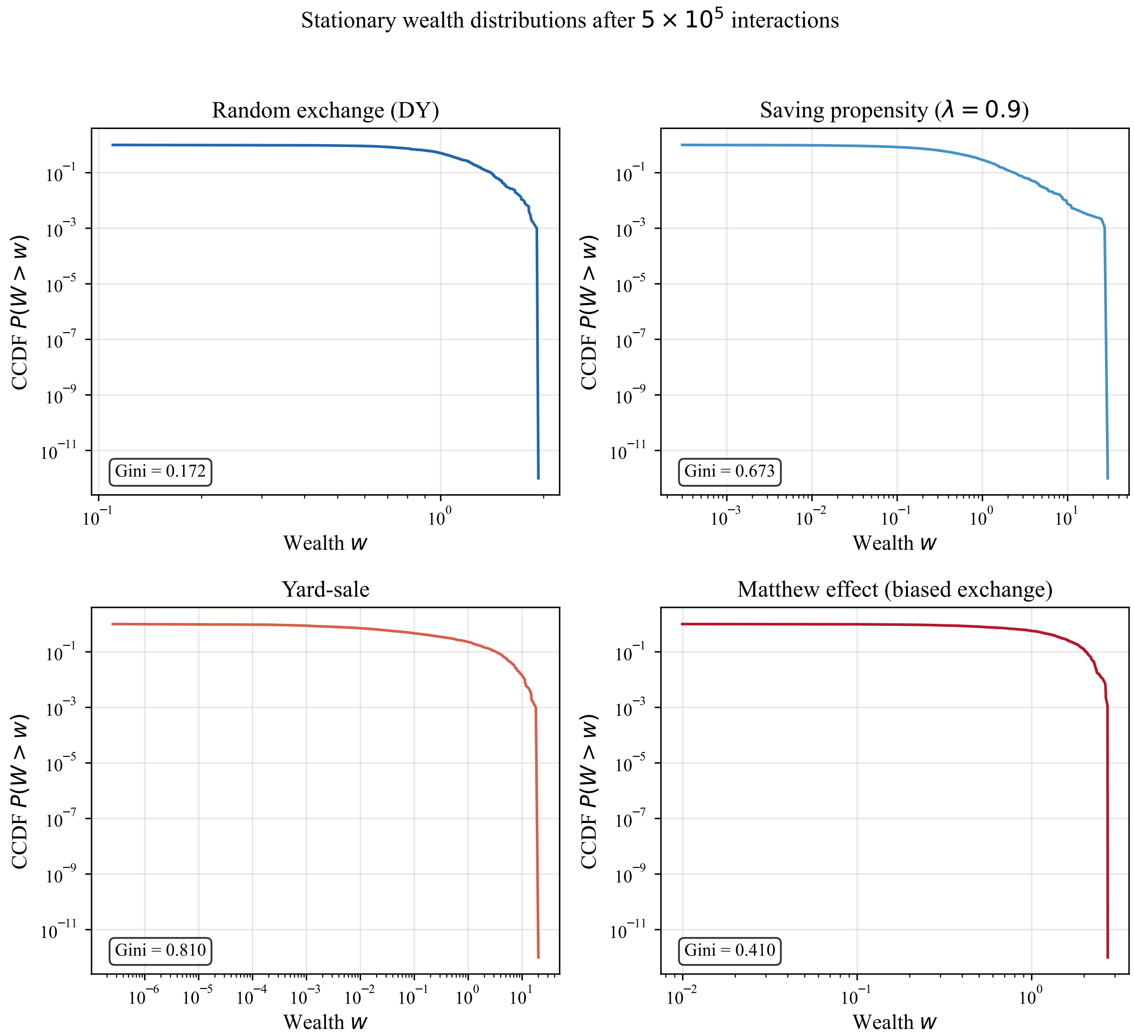

Figure 2: Terminal wealth distributions (CCDF) by model. Horizontal axes: agent wealth; vertical axes: fraction of agents with wealth exceeding w.

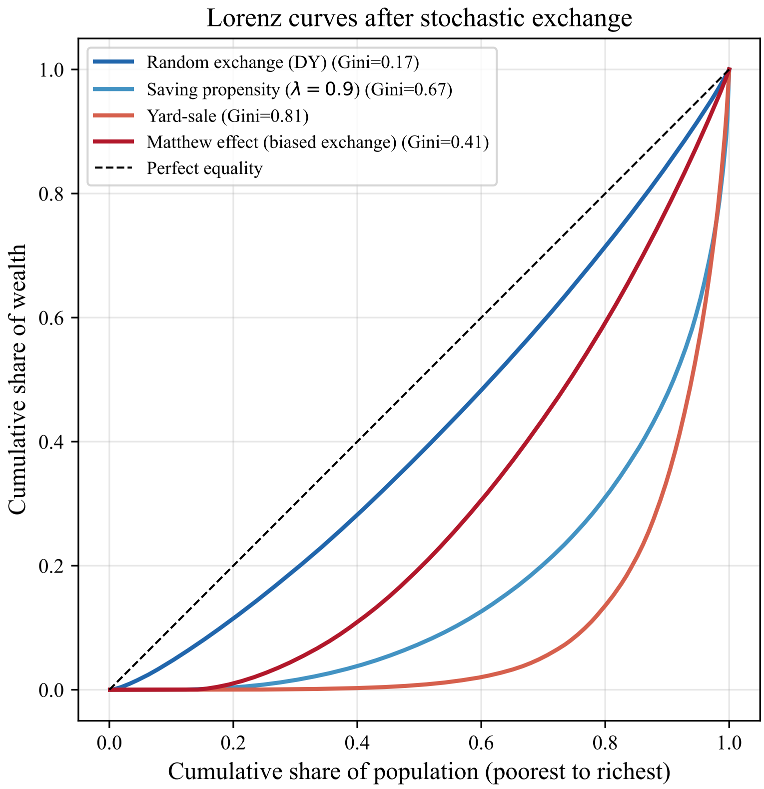

Figure 3: Lorenz curves at $t = 5 \times 10^5$. The diagonal is perfect equality.

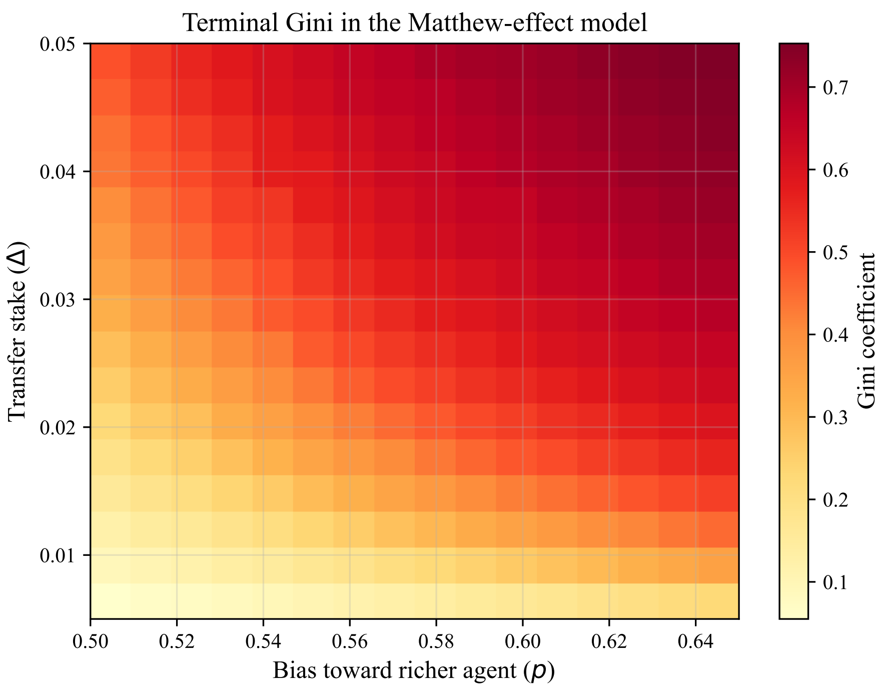

Figure 4: Terminal Gini in the Matthew-effect model as a function of transfer stake Δ and bias p toward the richer agent.

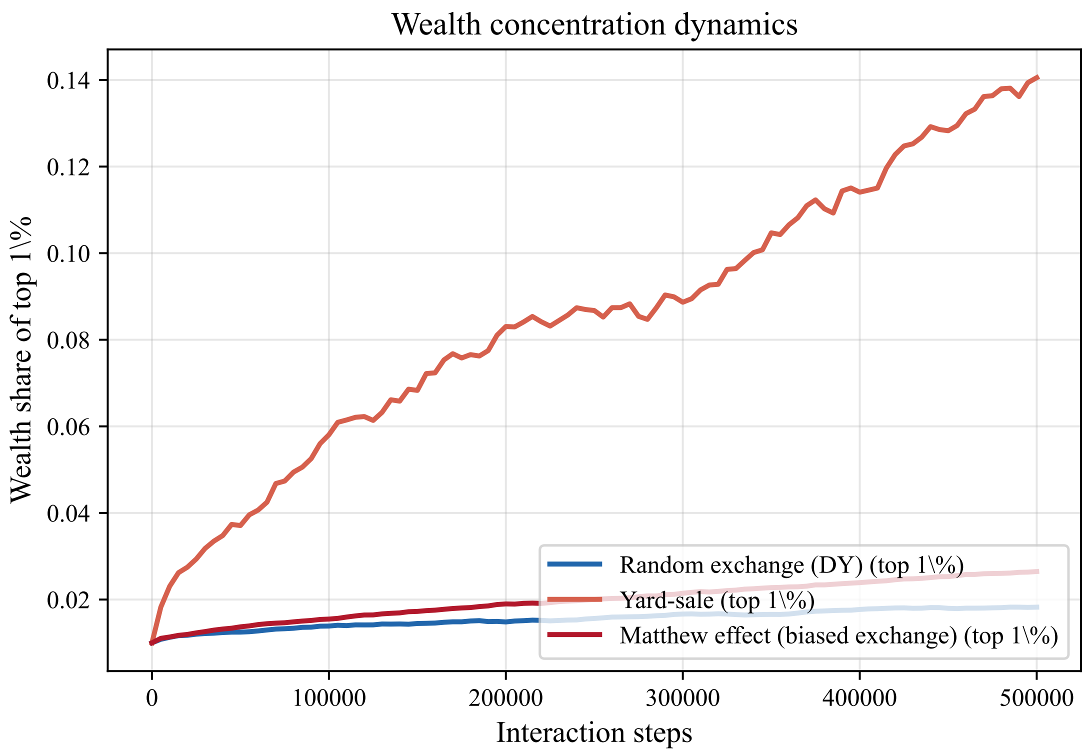

Figure 5: Top 1% wealth share over time for selected models.

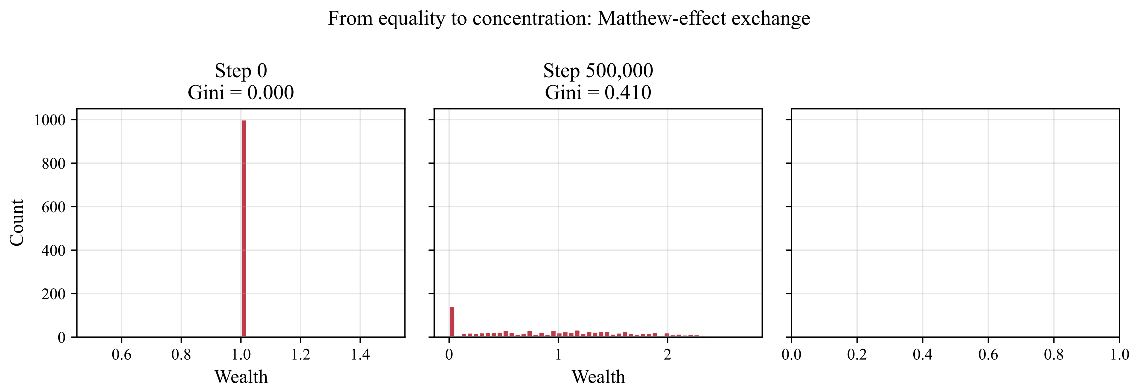

Figure 6: Wealth histograms at three time points under Matthew-effect exchange ($p = 0.55$).

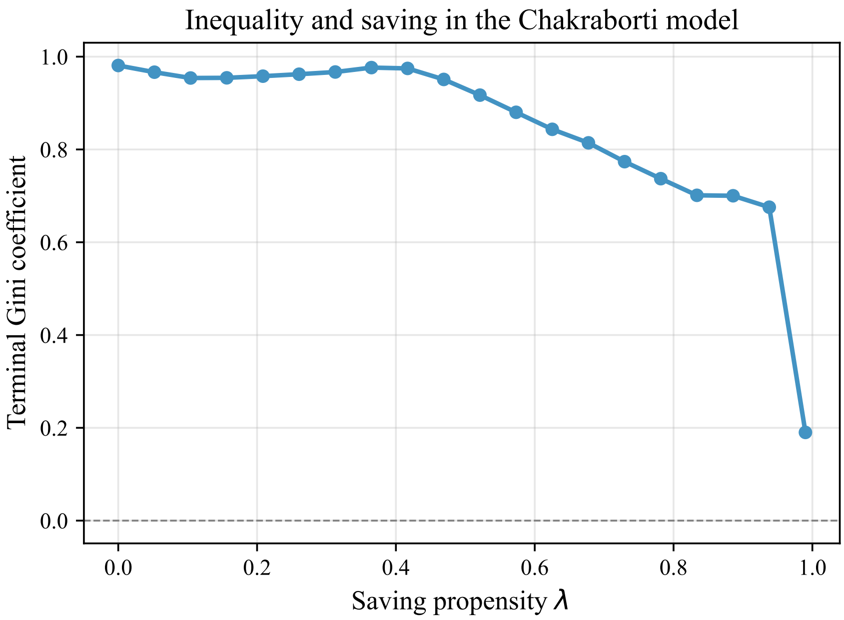

Figure 7: Terminal Gini as a function of saving propensity λ (Chakraborti model).

less produce dispersion because variance accumulates over time—an observation familiar from statistical mechanics (Dragulescu and Yakovenko, 2000).

Second, **multiplicative updates**: proportional losses (yard-sale) make poor agents disproportionately vulnerable. This is the mathematical core of wealth condensation (Bouchaud and Mézard, 2000).

Third, **biased accumulation**: even weak preferential flow (p slightly above one half) tilts the long-run distribution toward concentration, echoing Merton's and Price's cumulative-advantage processes (Merton, 1968; Price, 1976).

For a lay reader: imagine a classroom where every student starts with $10 and pairs up each day to flip a coin for $1. After many weeks, some students will hold far more than $10 and others far less, even though the coin is fair—pure luck, accumulated. If the coin is slightly rigged so the richer student wins more often, divergence accelerates. That is the Matthew effect in miniature.

5.2 Relation to empirical inequality

Real-world Gini coefficients for wealth commonly exceed 0.7 in high-income countries; our minimalist exchange models can reach similar local values only in condensation regimes (yard-sale) or after extreme time horizons. Production economies differ along dimensions these models omit: skill-biased technical change, bequests, assortative matching, and policy (Piketty, 2014; Benhabib et al., 2019; Galor and Zeira, 1993). The ABM contribution is not to "fit" a country snapshot, but to prove existence: *interaction alone* can generate inequality from equality.

Agent-based modeling complements reduced-form econometrics here (Farmer and Foley, 2009). Regression coefficients on initial conditions identify correlates in data; ABMs display possible *dynamic paths* consistent with symmetric fundamentals. Neither replaces the other.

5.3 Limitations and identification

We stress three cautions aligned with quantitative best practice.

1. **Stylized mechanisms, not causal claims about real policy.** Raising p in simulation raises Gini; that comparative static is *internally valid* in the model. Claiming that a specific real-world institution maps to $p = 0.55$ would require separate identification.

2. **Finite-N and stopping time.** Yard-sale condensation is asymptotic; finite runs approximate but do not fully reach the degenerate state.

3. **Conservation vs. growth.** Closed-exchange models conserve total wealth; growing economies with capital accumulation require extensions (Benhabib and Bisin, 2011).

6 Conclusion

Identical starting points do not guarantee identical outcomes. In agent-based wealth-exchange models with conserved aggregate resources, stochastic pairwise interaction generates inequality endogenously. The Matthew effect—a slight bias favoring those who already have more—accelerates concentration, but even fair random exchange suffices to destroy perfect equality over time.

The epigraph from Matthew captures a mechanism, not a moral theorem. Formalizing that mechanism in transparent ABMs clarifies which features of inequality require initial heterogeneity and which arise from the geometry of luck compounded through social and economic encounter (Merton, 1968; Yakovenko and Rosser, 2009; Ball, 2004). Future work might embed exchange kernels in networks (Di Matteo et al., 2001), calibrate λ and β to micro transaction data, or couple agents to production and fiscal policy in the tradition of Benhabib and Bisin (2011) and Piketty (2014).

References

Acemoglu, D. and Robinson, J. A. (2015). The rise and decline of general laws of capitalism. *Journal of Economic Perspectives*, 29(1):3–28.

Ball, P. (2004). *Critical Mass: How One Thing Leads to Another*. Farrar, Straus and Giroux, New York.

Benhabib, J. and Bisin, A. (2011). The distribution of wealth and fiscal policy in economies with finitely lived agents. *Econometrica*, 79(1):123–157.

Benhabib, J., Bisin, A., and Lisi, S. (2019). Wealth distribution and social mobility in the united states: 1940–2020. *The Quarterly Journal of Economics*, 134(3):1403–1458.

Bouchaud, J.-P. and Mézard, M. (2000). Wealth condensation in a simple model of economy. *Physica A: Statistical Mechanics and its Applications*, 282(3–4):536–545.

Chakraborti, A. and Chakrabarti, B. K. (2000). Statistical mechanics of money: How saving propensity affects its distribution. *The European Physical Journal B*, 17:167–170.

Chatterjee, A., Chakrabarti, B. K., and Manna, S. S. (2004). Pareto law in a kinetic model of market with random saving propensity. *Physica A: Statistical Mechanics and its Applications*, 335(1–2):155–163.

Di Matteo, T., Aste, T., and Hyde, S. T. (2001). Exchanges in complex networks and wealth distribution. *The European Physical Journal B*, 22:235–240.

Dragulescu, A. and Yakovenko, V. M. (2000). Statistical mechanics of money. *The European Physical Journal B*, 17:723–729.

Epstein, J. M. (2006). Generative social science: Studies in agent-based computational modeling.

Farmer, J. D. and Foley, D. (2009). The economy needs agent-based modelling. *Nature*, 460:685–686.

Gabaix, X. (2009). Power laws in economics and finance. *Annual Review of Economics*, 1:255–294.

Galor, O. and Zeira, J. (1993). Income distribution and macroeconomics. *The Review of Economic Studies*, 60(1):35–52.

Gini, C. (1912). Variabilità e mutabilità.

Holy Bible, New Revised Standard Version (1989). Matthew 25:29. "For to all those who have, more will be given, and they will have an abundance; but from those who have nothing, even what they have will be taken away.".

Ispolatov, S., Krapivsky, P. L., and Redner, S. (1998). Wealth distributions in asset exchange models. *The European Physical Journal B*, 2:267–276.

Lorenz, M. O. (1905). Methods of measuring the concentration of wealth. *Publications of the American Statistical Association*, 9(70):209–219.

Lux, T. (2008). Applications of statistical physics in finance and economics. *Physics of Complex Systems*, pages 685–729. Chapter in *Econophysics and Sociophysics: Trends and Perspectives*, Wiley-VCH.

Merton, R. K. (1968). The Matthew effect in science. *Science*, 159(3810):56–63.

Piketty, T. (2014). *Capital in the Twenty-First Century.* Harvard University Press, Cambridge, MA.

Price, D. J. d. S. (1976). A general theory of bibliometric and other cumulative advantage processes. *Journal of the American Society for Information Science*, 27(5–6):292–306.

Schelling, T. C. (1971). Dynamic models of segregation. *Journal of Mathematical Sociology*, 1(2):143–186.

Yakovenko, V. M. and Rosser, J. Barkley, J. (2009). Colloquium: Statistical mechanics of money, wealth, and income. *Reviews of Modern Physics*, 81:1703–1725.